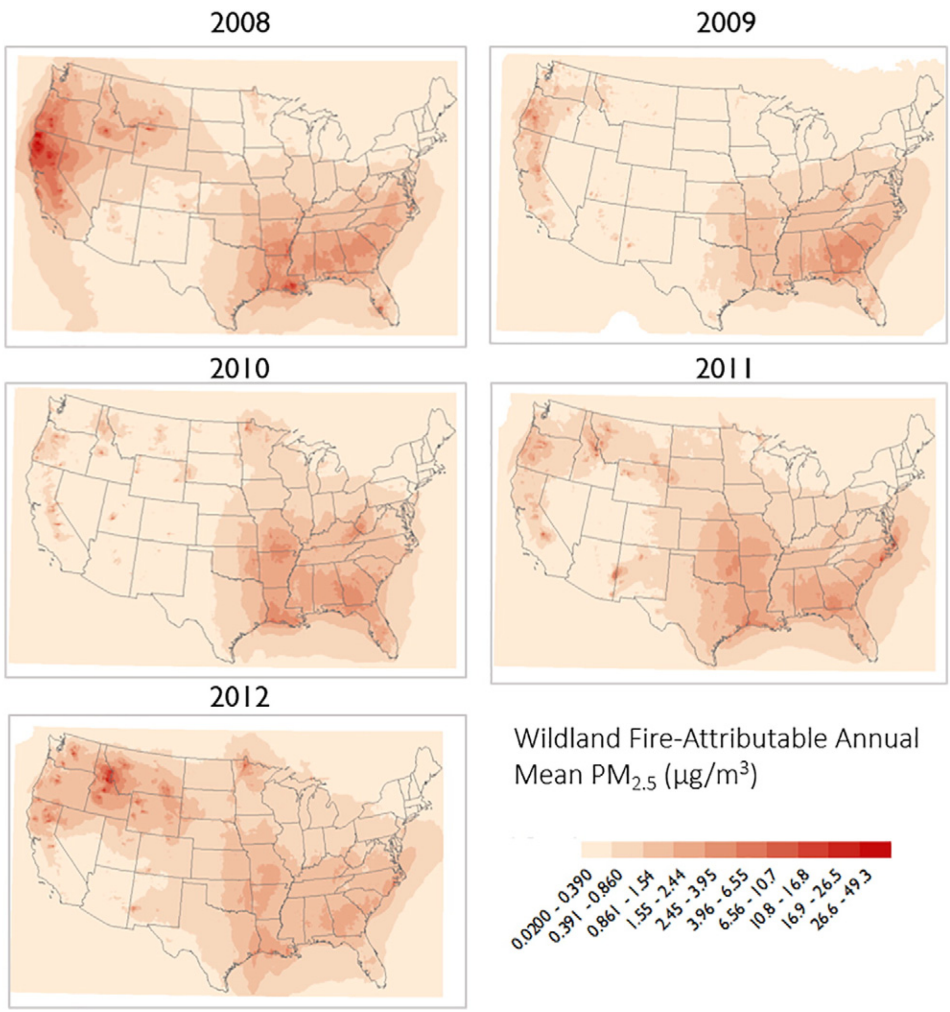

Fig. 1. Annual mean wildland fire-attributable $PM_{2.5}$ concentrations (2008–2012).

particles during wildfires restrict air flow to the pumps, shutting the monitors down and missing high concentration episodes, thus leading to potentially biased observed values. We also recognize the potential for CMAQ to overestimate wildland fire impacts (Baker et al., 2016). More complex calibration and data fusion models specific to the wildland fires are likely to become available with the increased usability of remote sensing in this area of research as well as improved parameterizations based on results from field campaigns such as the Fire and Smoke Model Evaluation Experiment (FASMEE) (http://www.fasmee.net).

Table 2
Premature deaths and illnesses attributable to wildfire-related $PM_{2.5}$ concentrations in each year calculated using alternative concentration-response functions (95% confidence intervals).

Endpoint	Year[a]				
	2008	2009	2010	2011	2012
Respiratory hospital admissions					
Delfino et al. (2009)	8500 (4400–12,000)	5200 (2700–7700)	6200 (3200–9100)	6300 (3300–9300)	6400 (3300–9400)
Zanobetti et al. (2009)	6300 (3600–9000)	3900 (2300–5500)	4600 (2600–6500)	4700 (2700–6700)	4800 (2800–6800)
Cardiovascular hospital admissions					
Delfino et al. (2009)	2800 (−500–6000)	1700 (−320–3700)	2100 (−380–4400)	2100 (−380–4500)	2100 (−390–4600)
Premature deaths from short-term exposure to $PM_{2.5}$					
Zanobetti and Schwartz (2009)	2500 (1900–3000)	1500 (1100–1800)	1700 (1300–2100)	1900 (1400–2200)	1800 (1400–2200)
Premature deaths from long-term exposure to $PM_{2.5}$					
Krewski et al. (2009)	14,000 (9700–19,000)	8700 (5800–11,000)	10,000 (6900–14,000)	11,000 (7300–14,000)	11,000 (7600–15,000)
Lepeule et al. (2012)	32,000 (16,000–48,000)	19,000 (9800–29,000)	23,000 (12,000–35,000)	24,000 (12,000–36,000)	25,000 (13,000–38,000)

[a] Values rounded to two significant figures; all functions estimated for populations ages 0–99.

Table 3
Estimated economic value of wildfire-attributable $PM_{2.5}$-related premature deaths and respiratory hospital admissions (2008 to 2012) (billions of 2010$, 95% confidence intervals).[a]

Health endpoints	Year					Present value
	2008	2009	2010	2011	2012	
Sum of mortality from short-term exposures and respiratory hospital admissions[a]	$20 ($2–$53)	$12 ($1–$31)	$14 ($1–$37)	$11 ($1–$30)	$12 ($1–$31)	$63 ($6–$170)
Sum of mortality from long-term exposures and respiratory hospital admissions[b]	$130 ($12–$340)	$76 ($7–$210)	$90 ($8–$250)	$96 ($9–$260)	$100 ($9–$270)	$450 ($42–$1000)

[a] Sum of Delfino et al. (2009) respiratory hospital admission estimates and Zanobetti and Schwartz (2009) mortality.
[b] Sum of Delfino et al. (2009) hospital admission estimates and Krewski et al. (2009) mortality.

The epidemiological literature reporting risks from wildland fire-related $PM_{2.5}$ is sparse; this makes quantifying wildland-attributable risks challenging. The epidemiology studies that the U.S. EPA and others commonly use to quantify PM-related mortality and morbidity impacts, and reported in the Supplemental materials to this article, were not explicitly designed to characterize risks from this emission source. However, it is reasonable to expect that many epidemiologic studies that did not explicitly address wildland fire impacts will have at least partially accounted for them, given that wildland fire particles tend to account for large portion of PM in the atmosphere (Verma et al., 2014). The National Emissions inventory of 2011 estimates that 41% of $PM_{2.5}$ emissions originate with wildland burning (U.S. EPA, 2011).

In light of this limitation, when we designed the health impact assessment, we took two steps to ensure that we were using effect coefficients that were well matched to the unique characteristics of wildland fire smoke events. First, we performed a random effects quantitative meta-analysis of respiratory hospital admission epidemiological studies of smoke events in the U.S. and elsewhere in the world. Second, we selected concentration-response relationships from epidemiological studies that we believe would better account for the episodic nature of smoke events and the corresponding elevated levels of particles (Zanobetti et al., 2009; Zanobetti and Schwartz, 2009).

The overall economic value of wildland fire-attributable premature deaths and respiratory hospital admissions is considerable. Depending on whether we quantify short-term $PM_{2.5}$-related premature deaths or long-term $PM_{2.5}$-realted deaths, the cumulative 5-year economic value is in the tens to hundreds of billions of dollars.

Nearly all states in the U.S. experienced elevated fine particle concentrations from wildland fire events over the 5-year period, according to our simulation of air quality. Certain states were affected by severe wildland fire events occurring across two or more years, including Louisiana (some of which may be due to debris burning after an active hurricane season because remote sensing of fires cannot distinguish between large debris burning from wildfires or prescribed fires), California, Idaho, and Georgia. Within these states, certain population subgroups were affected disproportionately. In particular, black populations, and to a lesser extent Asian populations, accounted for a greater share of wildland fire-attributable health impacts. State and local officials may wish to consider how best to communicate with subgroups that are most affected when deploying warning systems that alert the public to the health risks of wildland fire smoke. State and local officials deploying warning systems to alert the public to the health risks of wildland fire smoke may wish to consider how best to reach these subgroups.

This analysis is subject to certain limitations and uncertainties that shape the way in which the results may be interpreted. Considering first the air quality inputs, we assessed the performance of the chemical transport model by matching CMAQ grid locations to the locations of environmental monitors and comparing predicted and observed values. We found that the model is biased high when predicting low levels of

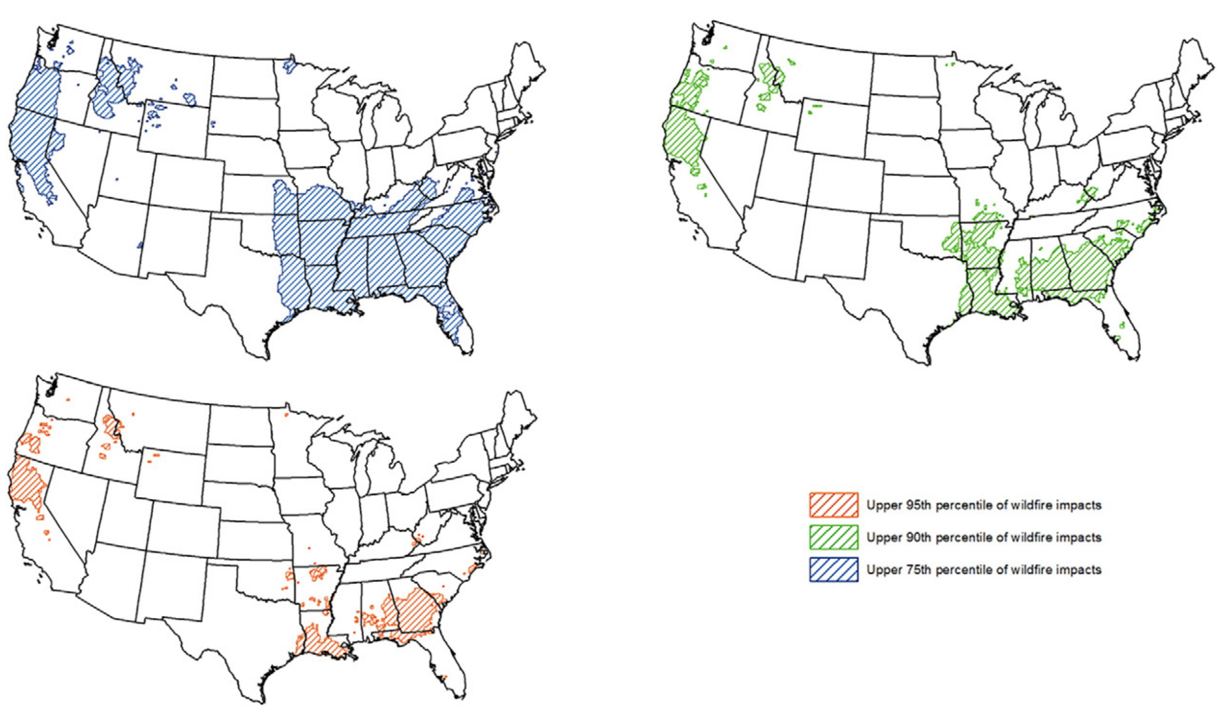

Fig. 2. Locations of the U.S. experiencing elevated wildfire-related $PM_{2.5}$ concentrations over a 5-year period.

Table 4
The percentage of individuals living in locations highly affected, and less affected, by wildland fires (2008 to 2012).[a]

Race	National average	Location of the U.S.[b]		Difference between highly and less affected
		Highly affected	Less affected	
Asian	5%	4%	5%	−1%
Black	13%	18%	13%	12%
Native American	1%	2%	1%	1%
White	81%	75%	81%	−6%
Total	100%	100%	100%	

[a] Highly affect subgroups are those individuals who have experienced cumulative levels of wildland fire attributable $PM_{2.5}$ concentrations that are at or above the 75th percentile, identified in Fig. 2 with blue hatched shading. Less affected subgroups are those not living in these areas.
[b] Estimates rounded to two significant figures.

$PM_{2.5}$ concentrations. The model also over predicts $PM_{2.5}$ (mainly Organic Carbon and Elemental Carbon) during all seasons for fire events. Baker et al. (2016) found the Flint Hills fire overestimated OC and EC but the Wallow fire did not overestimate to the same degree suggesting there is a lot of complexity in terms of how different fires are characterized in CMAQ. Another limitation is that only fire events that are part of the emission inventory have been evaluated. Any misspecification of emissions in the inventory fires is not included in our analysis. Further scientific advances and research efforts aimed to improve characterization of fire, fuel and emission inventories, and to improve measure characterization of differential toxicity of smoke emitted from different fuel types can potentially improve future health risk assessments.

The health impacts quantified using concentration-response relationships described in this paper are also subject to uncertainties. More specifically, questions remain regarding the extent to which certain species of fire-attributable PM may be more or less toxic than others (Sullivan et al., 2008). Thus, using effect coefficients from the epidemiological studies that did not specifically consider wildland fire episodes, may under- or over-estimate impacts. By contrast, the quantitative meta-analysis draws upon epidemiological studies of wildland fire events, but is limited to respiratory hospital admissions and PM_{10}. Several of these studies were conducted outside of the U.S.; differences between the health care system, wildland fire particle composition, behavioral responses to smoke events and other factors may bias the meta-analysis pooled risk coefficient. We also apply the effect coefficients from the Delfino et al. (2009) study of wildfires in Southern California nationwide; this may bias our estimates of risk high or low depending on the area that it is applied to. Finally, the present value calculation assumes that the wildland fire-related premature deaths estimated for each year are independent of those estimated in all other years. For example, individuals modeled as dying prematurely from wildland fire smoke exposure in year 1 cannot also die in year 2, but the net present value calculation introduces the possibility that deaths may have been counted more than once across years. However, given that the fraction of total deaths accounted for by air pollution episodes is relatively small (approximately 6% on a national basis), the potential for this approach to bias-high the number of wildland-attributable deaths is likely small (Fann et al., 2011).

Despite these uncertainties, this manuscript also exhibits a number of strengths and unique features. First, this analysis is, to our knowledge, the first to characterize the incidence and economic value of human health impacts attributable to wildland fire-emitted fine particles in the continental U.S. over a 5-year period. Second, while previous studies have incorporated a systematic review of wildland fire studies, none to our knowledge have performed a quantitative meta-analysis. Third, we characterize both the size, and distribution, of wildland fire-related premature deaths and hospital admissions across population subgroups. Taken together, the results in this analysis suggest that the number and value of wildland fire events is considerable and that these impacts are not shared equally across the U.S. population.

Disclaimer

The research described in this article has been reviewed by the National Health and Environmental Effects Research Laboratory, U.S. Environmental Protection Agency, and approved for publication. Approval does not signify that the contents necessarily reflect the views and policies of the Agency, nor does the mention of trade names of commercial products constitute endorsement or recommendation for use.

Competing financial interests

The authors declare no competing financial interests.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at http://dx.doi.org/10.1016/j.scitotenv.2017.08.024.

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